BABU BANARASI DAS UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

StudentManagementSystemUsing MongoDB

Submitted to:

Mr. HarendraSingh

Department:BCA (DS &AI)

Date: 25th September

Submitted by:

MOHD KAIF MERAZ

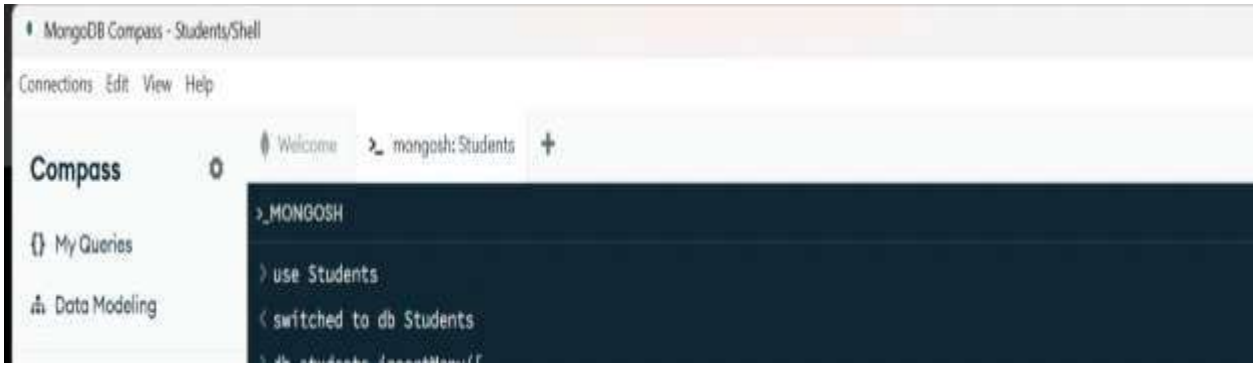
Roll no: (1250258273

1. CREATE DATABASE

MongoDB Query:

```
use Students
```

Explanation: Creates or switches to the Student database.

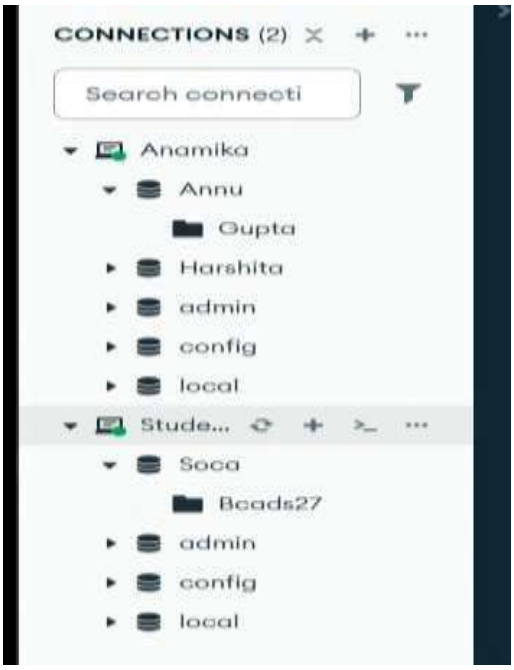


2. CREATE COLLECTION

MongoDB Query:

db. create Collection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

db. students. Insert Many([ ])

{roll:1, name:"Anamika", age:19,marks:88,city:"Lucknow"},

{roll:2, name:"Uday", age:18,marks:76,city:"Jaunpur"},

{roll:3, name:"Vaishnavi", age:19,marks:92,city:"Kanpur"},

{roll:4, name:"Annu", age:18,marks:69,city:"Ayodhya"},

{roll:5, name:"Vansh", age:19,marks:81,city:"Varanasi"},

{roll:6, name:"Amrita", age:20,marks:85,city:"Lucknow"},

{roll:7, name:"Naveen", age:19,marks:73,city:"Prayagraj"},

{roll:8, name:"Pratik", age:19,marks:90,city:"Lucknow"},

{roll:9, name:"Sangita", age:20,marks:78,city:"Gorakhpur"},

{roll:10, name:"Vivek", age:18,marks:65,city:"Kanpur"},

{roll:11, name:"Tarkeshwar", age:20,marks:71,city:"Jaunpur"},

{roll:12, name:"Shreya", age:19,marks:95,city:"Lucknow"},

{roll:13, name:"Akshita", age:18,marks:84,city:"Ayodhya"},

{roll:14, name:"Shruti", age:19,marks:87,city:"Prayagraj"},

{roll:15, name:"Kartik", age:20,marks:74,city:"Varanasi"},

{roll:16, name:"Sejal", age:18,marks:89,city:"Lucknow"},

{roll:17, name:"Astha", age:19,marks:82,city:"Kanpur"},

{roll:18, name:"Jai", age:18,marks:58,city:"Gonda"},

{roll:19, name:"Harshita", age:20,marks:91,city:"Lucknow"},

{roll:20, name:"Omkar", age:19,marks:63,city:"Bareilly"},

{roll:21, name:"Ayush", age:18,marks:77,city:"Lucknow"},

{roll:22, name:"Saumya", age:19,marks:86,city:"Jaunpur"},

{roll:23, name:"Vidya", age:20,marks:72,city:"Kanpur"},

{roll:24, name: "Adarsh", age:19,marks:80,city:"Ayodhya"},

{roll:25, name: "Anmol", age:18,marks:67,city:"Prayagraj"},

{roll:26, name: "Anushka", age:20,marks:94,city:"Lucknow"},

{roll:27, name: "Ananya", age:19,marks:88,city:"Varanasi"},

{roll:28, name: "Bhaskar", age:18,marks:61,city:"Gorakhpur"},

{roll:29, name: "Rohan", age:19,marks:75,city:"Kanpur"},

{roll:30, name: "Nishant", age:20,marks:79,city:"Lucknow"},

{roll:31, name: "Vineet", age:18,marks:68,city:"Jaunpur"},

{roll:32, name: "Anika", age:19,marks:90,city:"Lucknow"},

{roll:33, name: "Jai", age:20,marks:55,city:"Ayodhya"},

{roll:34, name: "Lakshya", age:18,marks:83,city:"Kanpur"},

{roll:35, name: "Krishna", age:19,marks:96,city:"Prayagraj"},

{roll:36, name: "Rumana", age:20,marks:70,city:"Lucknow"},

{roll:37, name: "Priyanshu", age:18,marks:81,city:"Varanasi"},

{roll:38, name: "Aditya", age:19,marks:66,city:"Gorakhpur"},

{roll:39, name: "Dhruv", age:20,marks:93,city:"Lucknow"},

{roll:40, name: "Nivedita", age:19,marks:85,city:"Kanpur"},

{roll:41, name: "Meenu", age:18,marks:74,city:"Jaunpur"},

{roll:42, name: "Akanksha", age:20,marks:89,city:"Lucknow"},

{roll:43, name: "Sarika",age:19,marks:62,city:"Ayodhya"},

{roll:44, name: "Anchal",age:18,marks:78,city:"Prayagraj"},

{roll:45, name: "Umra", age:20,marks:84,city:"Lucknow"},

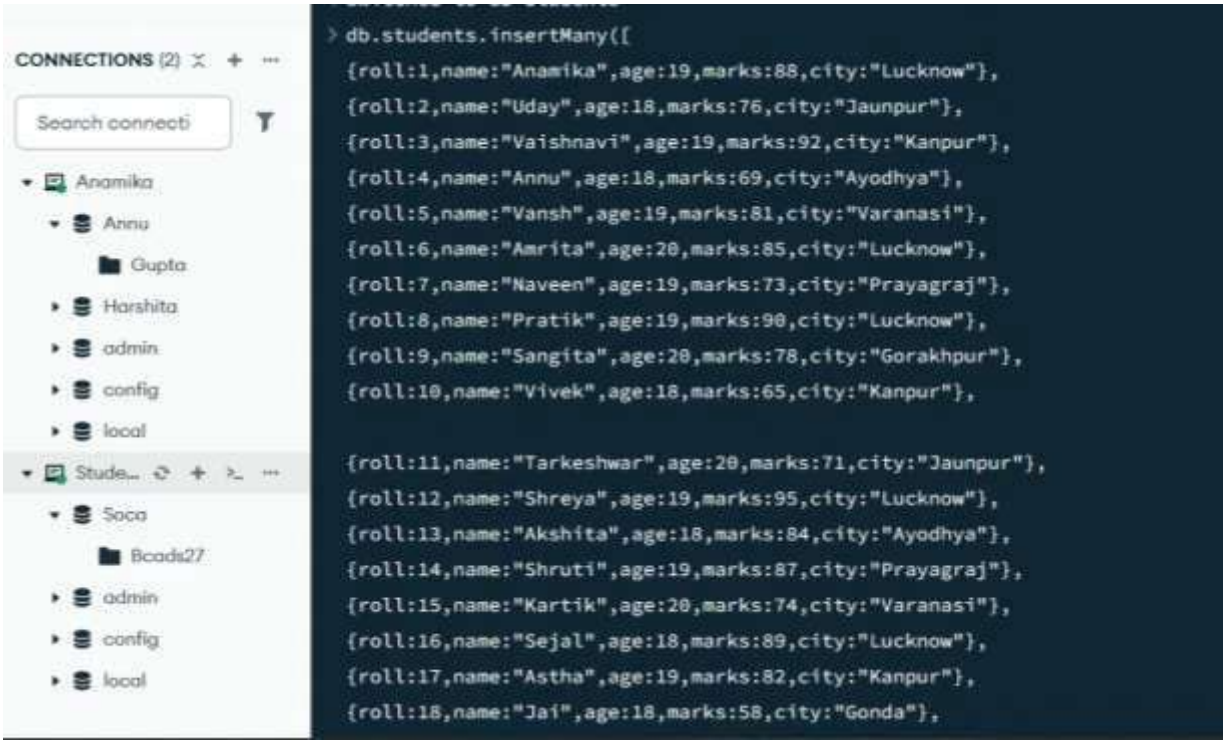
{roll:46, name: "Nashra", age:19,marks:73,city:"Kanpur"},

{roll:47, name: "Aman", age:18,marks:59,city:"Gonda"},

{roll:48, name: "Priya", age:20,marks:87,city:"Lucknow"},

{roll:49, name: "Rahul", age:19,marks:64,city:"Bareilly"},

{roll:50, name: "Neha", age:18,marks:91,city:"Varanasi"}



Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:
db. students. Find ()
Explanation: Retrieves documents from the students

collection.



5. EQUAL TO — $eq

MongoDB Query:

db. students. Find ({marks: {$eq: 88}})

Explanation: Finds students whose marks are exactly 88.

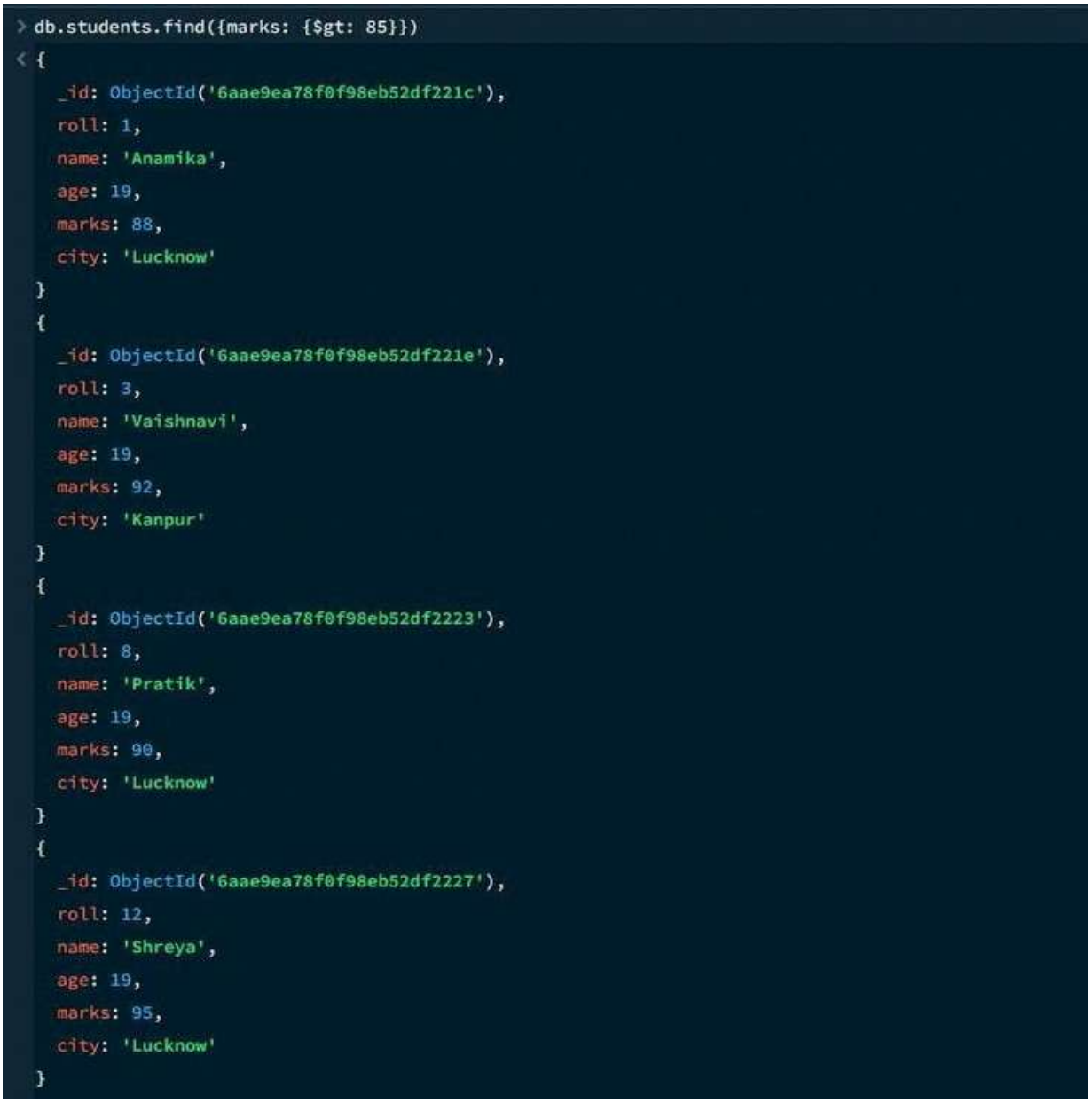


6. GREATER THAN — $gt

MongoDB Query:

db.students.find({marks: {$gt: 85}})

Explanation: Finds students whose marks are greater than 85.

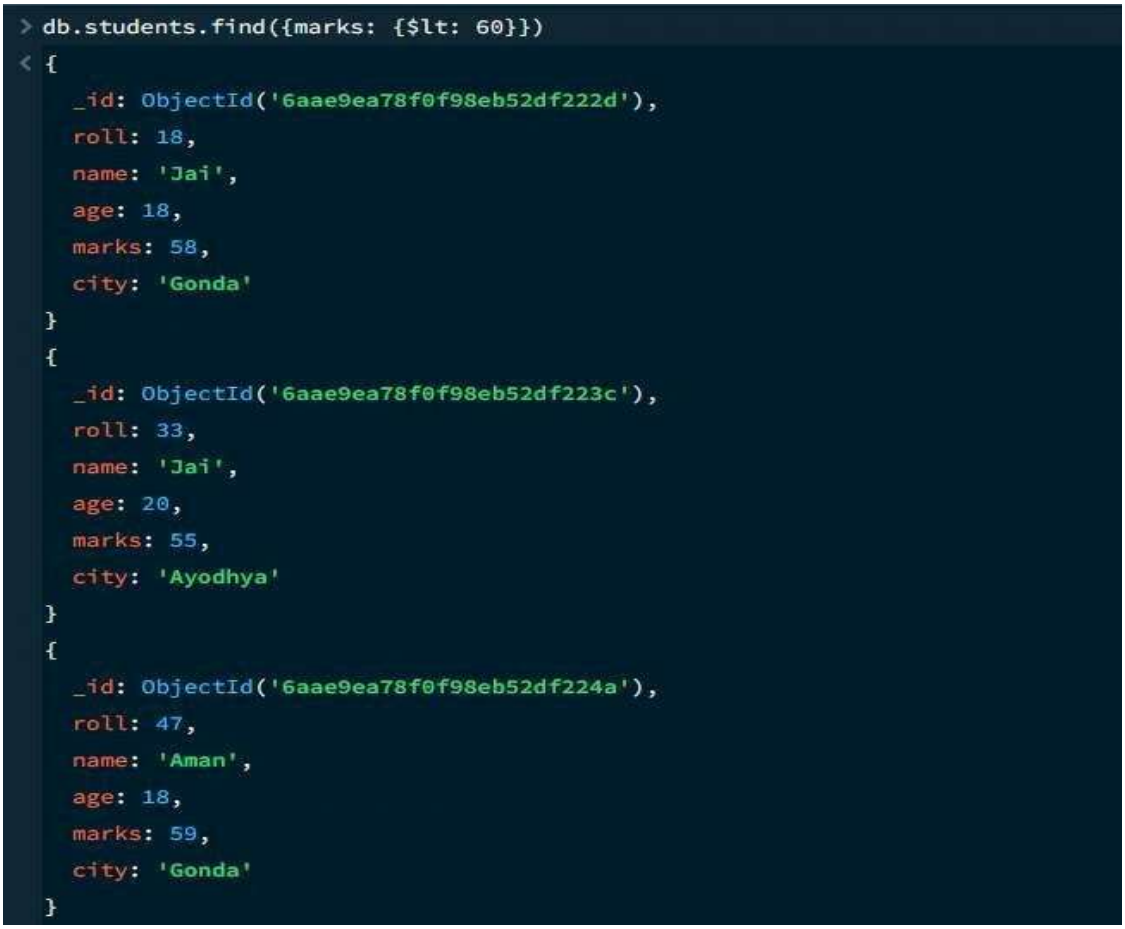


7. LESS THAN — $lt MongoDB

Query:

db. students. find ({marks: {$lt: 60}})

Explanation: Finds students whose marks are less than 60.

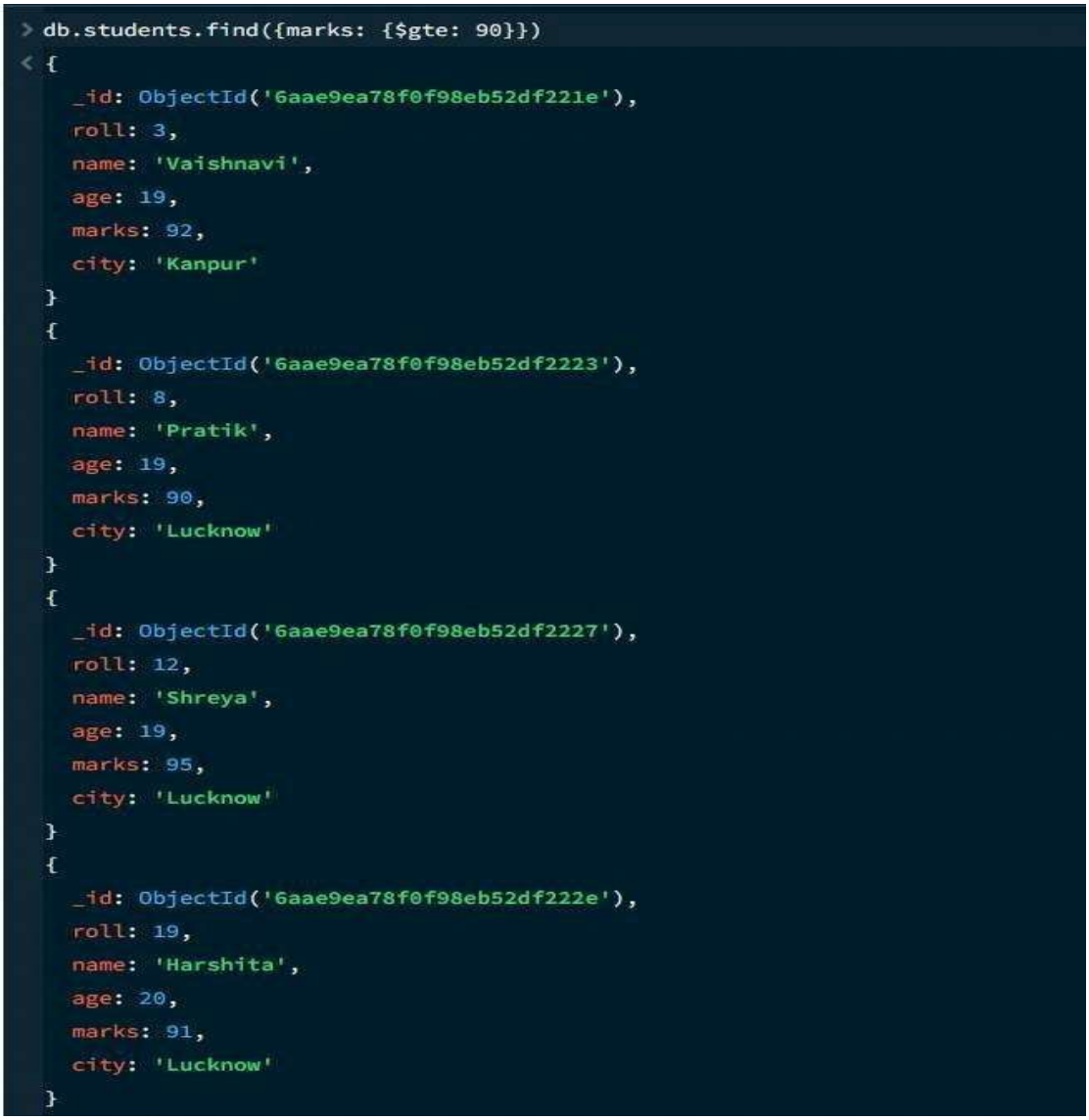


8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

db. students. find({marks: {$gte: 90}})

Explanation: Finds students whose marks are 90 or above.



```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
    marks: 90,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222e'),
    roll: 19,
    name: 'Harshita',
    age: 20,
    marks: 91,
    city: 'Lucknow'
  }
```

9. LESS THAN OR EQUAL TO

— $lte MongoDB Query:

db. students. find({marks: {$lte: 70}})

Explanation: Finds students whose marks are 70 or below.

```
> db.students.find({marks: {$lte: 70}})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221f'),
    roll: 4,
    name: 'Annu',
    age: 18,
    marks: 69,
    city: 'Ayodhya'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2225'),
    roll: 10,
    name: 'Vivek',
    age: 18,
    marks: 65,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222d'),
    roll: 18,
    name: 'Jai',
    age: 18,
    marks: 58,
    city: 'Gonda'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df222f'),
    roll: 20,
    name: 'Omkar',
    age: 19,
    marks: 63,
    city: 'Bareilly'
  }
```

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(

  {roll:5},
  {$set:{marks:95}}
)
```

Explanation: Updates the marks to 95 of the student with roll number 5.

```
> db.students.updateOne(
    {roll:5},
    {$set:{marks:95}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.students.find({roll:5})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df2220'),
    roll: 5,
    name: 'Vansh',
    age: 19,
    marks: 95,
    city: 'Varanasi'
```

11. DELETE ONE OPERATION MongoDB

Query:

db.students.deleteOne({roll:10})

Explanation: Deletes one document matching roll number 10.

```
  }
> db.students.deleteOne({roll:10})
< {
    acknowledged: true,
    deletedCount: 1
  }
> db.students.find({roll:10})
<
Students >
```

12. DELETE MANY

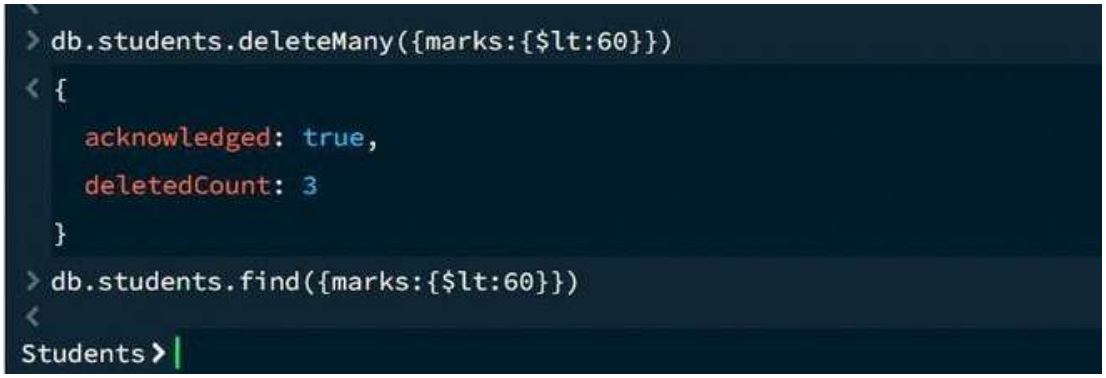
OPERATION MongoDB

Query:

```
db.students.deleteMany({

  marks: {$lt:60}
})
```

Explanation: Deletes all students having marks below 60.



13. AND OPERATION —

$and MongoDB Query:

```
db. students. find({

  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

```
>_MONGOSH
> db.students.find({
    $and:[
      {marks:{$gt:80}},{age:19}
    ]
  })
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
```

14. OR OPERATION — $or

MongoDB Query:

```
db.students.find({

  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
```

Explanation: Finds students from Lucknow or Kanpur.

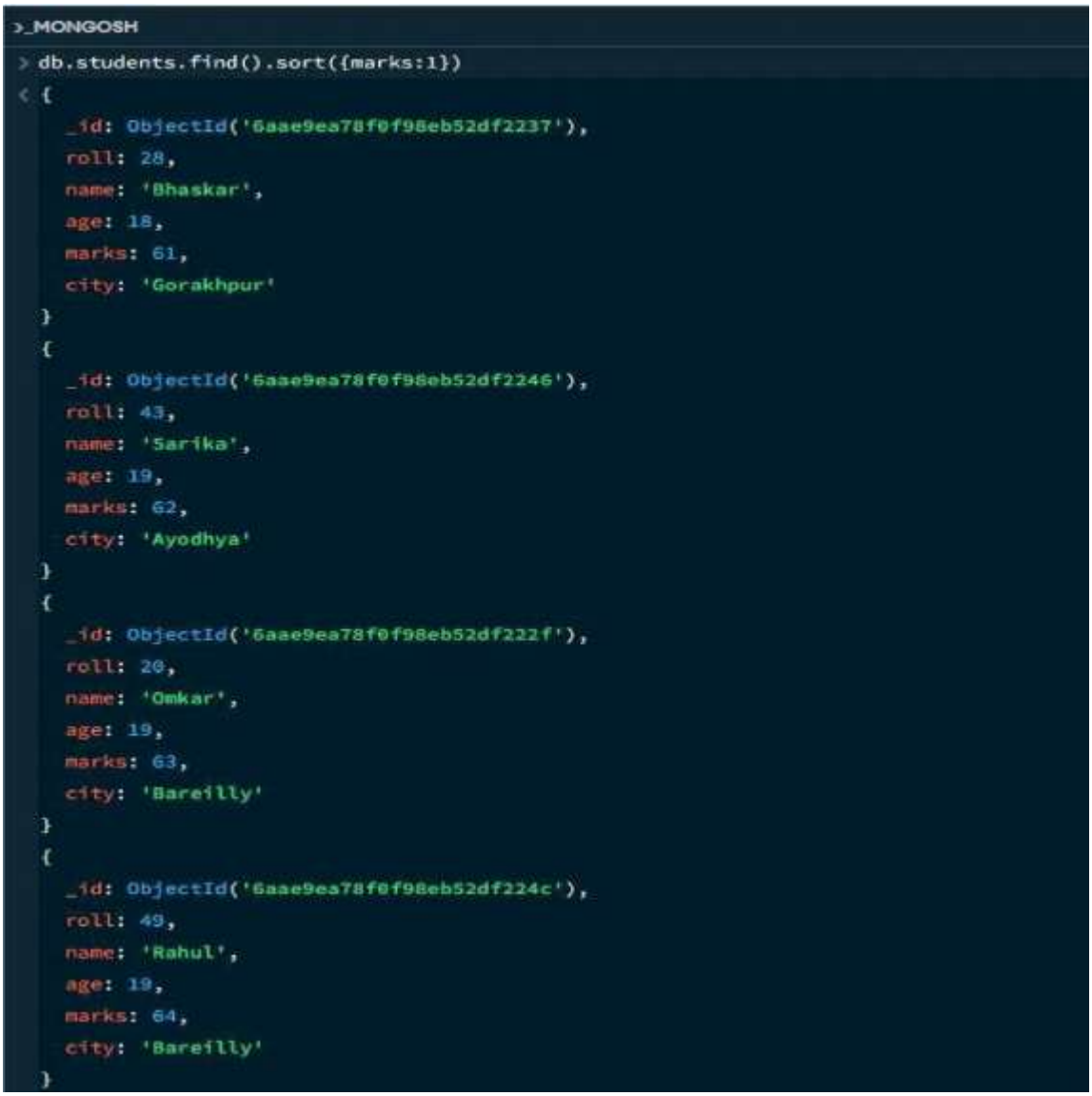
```
>_MONGOSH
> db.students.find({
    $or:[
      {city:"Lucknow"},{city:"Kanpur"}
    ]
  })
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
```

15. SORT OPERATION

A.MongoDB Query:

db. students. find (). sort({marks : 1})

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.



B. MongoDB Query:

db. students. find (). Sort ({marks: -1})

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
>_MONGOSH
> db.students.find().sort({marks:-1})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df223e'),
    roll: 35,
    name: 'Krishna',
    age: 19,
    marks: 96,
    city: 'Prayagraj'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2227'),
    roll: 12,
    name: 'Shreya',
    age: 19,
    marks: 95,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2220'),
    roll: 5,
    name: 'Vansh',
    age: 19,
    marks: 95,
    city: 'Varanasi'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2235'),
    roll: 26,
    name: 'Anushka',
    age: 20,
    marks: 94,
    city: 'Lucknow'
  }
```

16. LIMIT OPERATION

MongoDB Query:

db. students. find (). limit (5)

Explanation: Displays only the first five documents.

```
>_MONGOSH
> db.students.find().limit(5)
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221d'),
    roll: 2,
    name: 'Uday',
    age: 18,
    marks: 76,
    city: 'Jaunpur'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df221e'),
    roll: 3,
    name: 'Vaishnavi',
    age: 19,
```

17. SEARCH BY CITY

MongoDB Query:

db.students.find({city:"Lucknow"})

Explanation: Finds all students whose city is Lucknow.

```
>_MONGOSH
> db.students.find({city:"Lucknow"})
< {
    _id: ObjectId('6aae9ea78f0f98eb52df221c'),
    roll: 1,
    name: 'Anamika',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2221'),
    roll: 6,
    name: 'Amrita',
    age: 20,
    marks: 85,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6aae9ea78f0f98eb52df2223'),
    roll: 8,
    name: 'Pratik',
    age: 19,
```

18. COUNT DOCUMENTS

MongoDB Query:

db. students. count Documents ()

Explanation: Counts documents currently present in the collection (performing after the delete function).

```
>_MONGOSH
  }
> db.students.countDocuments()
< 46
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
$eq	Equal to	marks: {$eq:88}
$gt	Greater than	marks: {$gt:85}
$lt	Less than	marks: {$lt:60}
$gte	Greater than or equal to	marks: {$gte:90}
$lte	Less than or equal to	marks: {$lte:70}
$and	All conditions true	$and:[...]
$or	At least one condition true	$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as $eq, $gt, $lt, $gte, and $lte work, along with logical operators such as $and and $or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.

MONGODB CRUD OPERATIONS

1. CREATE operation:

```
db.students.insertMany([
  { student_id: 1, name:
    "Aarav Sharma", age: 19,
    gender: "Male", course:
    "BCA Data Science",
    semester: 3, city:
    "Lucknow", marks: 82
  },
  { student_id: 2, name:
    "Aditi Verma", age: 19,
    gender: "Female", course:
    "BCA Data Science",
    semester: 3, city:
    "Kanpur", marks: 88
  },
  { student_id: 3, name:
    "Aditya Singh", age:
    20, gender: "Male",
    course: "BCA AI",
    semester: 3, city:
    "Delhi", marks: 76
  },
  { student_id: 4, name:
    "Ananya Gupta", age:
    19,
    gender: "Female", course:
    "BCA AI",
    semester: 3, city:
    "Jaipur", marks:
    91
  },
  { student_id: 5, name:
    "Arjun Mehta", age: 20,
    gender: "Male", course:
    "BCA Data Science",
```

```
  semester: 3, city: "Agra",
  marks: 74
},
{ student_id: 6, name:
  "Bhavya Kapoor", age:
  19, gender: "Female",
  course: "BCA AI",
  semester: 3, city:
  "Delhi", marks: 86

},
{ student_id: 7, name:
  "Chirag Yadav", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 79
},
{ student_id: 8, name:
  "Diya Mishra", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
     "Varanasi",
  marks: 93
},
{ student_id: 9, name: "Dev
  Rajput", age: 20, gender:
  "Male", course: "BCA Data
  Science", semester: 3,
  city: "Meerut",
  marks: 81
},
```

```
{  student_id:  10,  name:
  "Ishita  Agarwal",  age:
  19,
  gender: "Female", course:
  "BCA AI",
  semester: 3,
  city: "Agra",
  marks: 89
},
{ student_id: 11, name:
  "Kabir Khan", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 72
},
{  student_id:  12,  name:
  "Kavya  Sharma",  age:
  19,
  gender:   "Female", course:
  "BCA AI",
  semester: 3, city:
  "Jaipur", marks:
  95
},
{
  student_id: 13, name:
  "Manav Jain", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 78
},
{ student_id: 14, name:
  "Mehak Sinha", age:
```

```
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
  "Delhi", marks:
  84
},
{ student_id: 15, name:
  "Naman Gupta", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city: "Agra",
  marks: 69
},
{ student_id: 16,
  name: "Navya
  Singh", age: 19,
  gender: "Female",
  course: "BCA AI",
  semester: 3, city:
  "Lucknow", marks:
  90
},
{ student_id: 17, name: "Om
  Prakash", age: 20, gender:
  "Male", course: "BCA Data
  Science", semester: 3,
  city: "Varanasi", marks:
  75
},
{ student_id: 18, name:
  "Palak Verma", age:
  19, gender:
     "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city:       "Kanpur",
  marks: 87
},
{ student_id: 19, name:
  "Raghav Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Jaipur", marks: 80
},
{
  student_id:      20,
  name: "Riya Gupta",
  age: 19, gender:
     "Female", course:
  "BCA AI", semester:
  3,
  city:       "Delhi",
  marks: 92
},
{ student_id: 21, name:
  "Rohan Singh", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 77
},
{  student_id:  22,  name:
  "Sakshi  Yadav",  age:
  19,
  gender:   "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city: "Agra",
  marks: 85
},
{ student_id: 23,
  name: "Samar
  Khan", age: 20,
  gender: "Male",
  course: "BCA Data
  Science",
  semester: 3, city:
  "Meerut", marks:
  73
},
{ student_id: 24, name:
  "Simran Kaur", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3, city:
  "Delhi", marks:
  94
},
{ student_id: 25, name:
  "Tanishq Jain", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 71
},
{ student_id: 26, name:
  "Tanvi  Mishra",  age:
  19,
  gender:   "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city: "Varanasi",
  marks: 89
},
{ student_id: 27, name:
  "Uday Singh", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 83
},
{  student_id:  28,  name:
  "Vanshika  Gupta",  age:
  19,
  gender: "Female", course:
  "BCA AI",
  semester: 3,
  city: "Jaipur",
  marks: 96
},
{ student_id: 29, name:
  "Varun Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city: "Agra",
  marks: 68
},
{
  student_id:
     30, name: "Zoya
  Khan", age: 19,
  gender: "Female",
  course: "BCA AI",
  semester: 3,
  city:      "Delhi",
```

```
  marks: 91
},
{ student_id: 31, name:
  "Abhishek Verma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Kanpur", marks: 79
},
{  student_id:  32,  name:
  "Anjali  Sharma",  age:
  19,
  gender:   "Female", course:
     "BCA   AI",
  semester: 3, city:
  "Lucknow", marks:
  88
},
{ student_id: 33,
  name: "Deepak
  Yadav", age: 20,
  gender: "Male",
  course: "BCA Data
  Science", semester: 3,
  city: "Varanasi", marks:
  74
},
{  student_id:  34,  name:
  "Garima  Singh",  age:
  19,
  gender:   "Female", course:
  "BCA AI",
```

```
  semester: 3,
  city: "Agra",
  marks: 90
},
{ student_id: 35, name:
  "Harsh Gupta", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Jaipur", marks: 82
},
{
  student_id:      36,
  name: "Isha Verma",
  age: 19, gender:
     "Female", course:
  "BCA AI", semester:
  3,
  city:       "Delhi",
  marks: 93
},
{ student_id: 37, name:
  "Karan Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 76
},
{  student_id:  38,  name:
  "Lavanya  Jain",  age:
  19,
  gender:   "Female", course:
  "BCA AI",
  semester: 3, city:
  "Kanpur", marks:
```

```
  87
},
{ student_id: 39, name:
  "Mohit Singh", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Meerut", marks: 70
},
{ student_id: 40,
  name: "Neha
  Gupta", age: 19,
  gender: "Female",
  course: "BCA AI",
  semester: 3, city:
  "Jaipur", marks:
  95
},
{ student_id: 41, name:
  "Pranav Kumar", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city: "Agra",
  marks: 81
},
{ student_id: 42, name:
  "Priya Singh", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3,
  city:      "Lucknow",
  marks: 89
},
```

```
{ student_id: 43, name:
  "Rahul Verma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3,
  city: "Delhi",
  marks: 73
},
{  student_id:  44,  name:
  "Ritika  Sharma",  age:
  19,
  gender:   "Female", course:
     "BCA   AI",
  semester: 3,
  city: "Kanpur",
  marks: 92
},
{ student_id: 45, name:
  "Shivam Gupta", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Varanasi", marks: 78
},
{ student_id: 46, name:
  "Shreya Jain", age:
  19, gender:
     "Female", course:
  "BCA AI",
  semester: 3,
  city: "Agra",
  marks: 86
},
{
```

```
  student_id: 47, name:
  "Varsha Singh", age: 19,
  gender: "Female", course:
  "BCA Data Science",
  semester: 3, city:
  "Lucknow", marks: 84
},
{ student_id: 48,
  name: "Vivek
  Yadav", age: 20,
  gender: "Male",
  course: "BCA AI",
  semester: 3, city:
  "Jaipur", marks: 80
},
{ student_id: 49, name:
  "Yash Sharma", age: 20,
  gender: "Male", course:
  "BCA Data Science",
  semester: 3, city:
  "Delhi", marks: 77
},
{ student_id: 50,
  name: "Muskan
  Verma", age: 19,
  gender: "Female",

      name: 'Manav Jain', age:
       20, gender: 'Male',
       course: 'BCA Data
       Science', semester: 3,
       city: 'Kanpur', marks: 78
     }

     {
       _id:  ObjectId('6aad46d1505f94ec0f315254'),
       student_id: 14, name: 'Mehak Sinha', age:
       19, gender: 'Female', course: 'BCA AI',
       semester: 3, city: 'Delhi', marks: 84
     }
```

```
{
  _id:
  ObjectId('6aad46d1505f94ec0f315255'),
  student_id: 15, name: 'Naman Gupta', age:
  20, gender: 'Male', course: 'BCA Data
  Science', semester: 3, city: 'Agra',
  marks: 69
}

{
  _id:  ObjectId('6aad46d1505f94ec0f315256'),
  student_id: 16, name: 'Navya Singh', age:
  19, gender: 'Female', course: 'BCA AI',
  semester: 3, city: 'Lucknow', marks: 90
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315257'),
  student_id: 17, name: 'Om Prakash', age:
  20, gender: 'Male', course: 'BCA Data
  Science', semester: 3, city: 'Varanasi',
  marks: 75
}

{
  _id:  ObjectId('6aad46d1505f94ec0f315258'),
  student_id: 18, name: 'Palak Verma', age:
  19, gender: 'Female', course: 'BCA AI',
  semester: 3, city: 'Kanpur', marks: 87
}

{
  _id:
  ObjectId('6aad46d1505f94ec0f315259'),
  student_id: 19, name: 'Raghav Sharma',
  age: 20, gender: 'Male', course: 'BCA
  Data Science', semester: 3, city:
  'Jaipur', marks: 80
}

{
  _id:  ObjectId('6aad46d1505f94ec0f31525a'),
  student_id: 20, name: 'Riya Gupta', age: 19,
  gender:  'Female',  course:  'BCA  AI',
  semester: 3, city: 'Delhi', marks: 92
}
```

```
db.students.findOne({student_id:10})
{
```

```
  _id: ObjectId('6aad46d1505f94ec0f315250'),
  student_id: 10, name: 'Ishita Agarwal',
  age: 19, gender: 'Female', course: 'BCA
  AI', semester: 3,
  city: 'Agra',
  marks: 89
}
```

3. UPDATE operation:

```
db.students.updateOne({student_id:1},{$set:{marks:90}})
{ acknowledged:
  true, insertedId:
  null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}

db.students.updateOne({students_id:5},{$set:{city:"Lucknow"}})
 { acknowledged:
  true, insertedId:
  null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
 }
 db.students.findOne({student_id:5})

 {
  _id:
   ObjectId('6aad46d1505f94ec0f31524b'),
   student_id: 5, name: 'Arjun Mehta', age:
   20, gender: 'Male', course: 'BCA Data
   Science', semester: 3, city: 'Agra',
```

```
  marks: 74
}
```

4. DELETE operation:

```
db.students.deleteOne({student_id: 40})

{ acknowledged:
true, deletedCount:
1 }



db.students.deleteMany({marks: {$lt: 70}})
{ acknowledged:
  true,
  deletedCount: 2
}
```